

Six Lessons I Learned on Giving Talks

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EC 2026 Mentoring Workshop



Lessons adapted from...



Mary



Moses



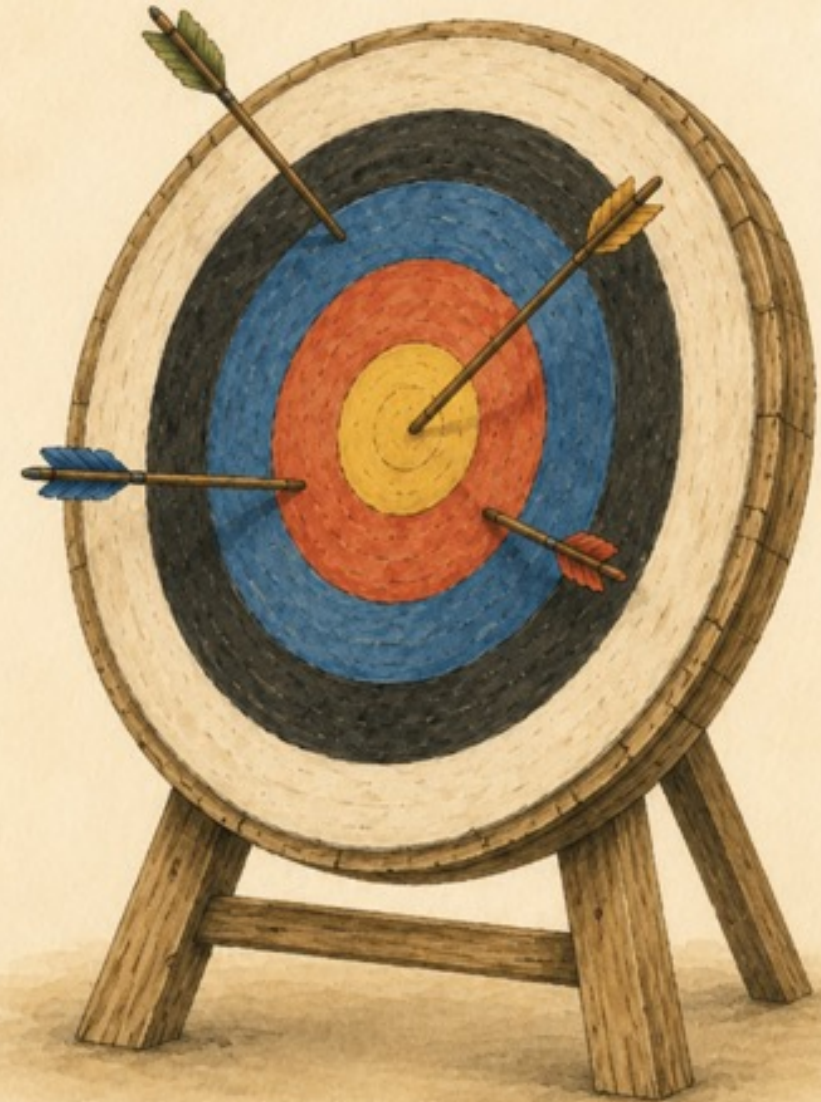
Li-Yang



Aviad

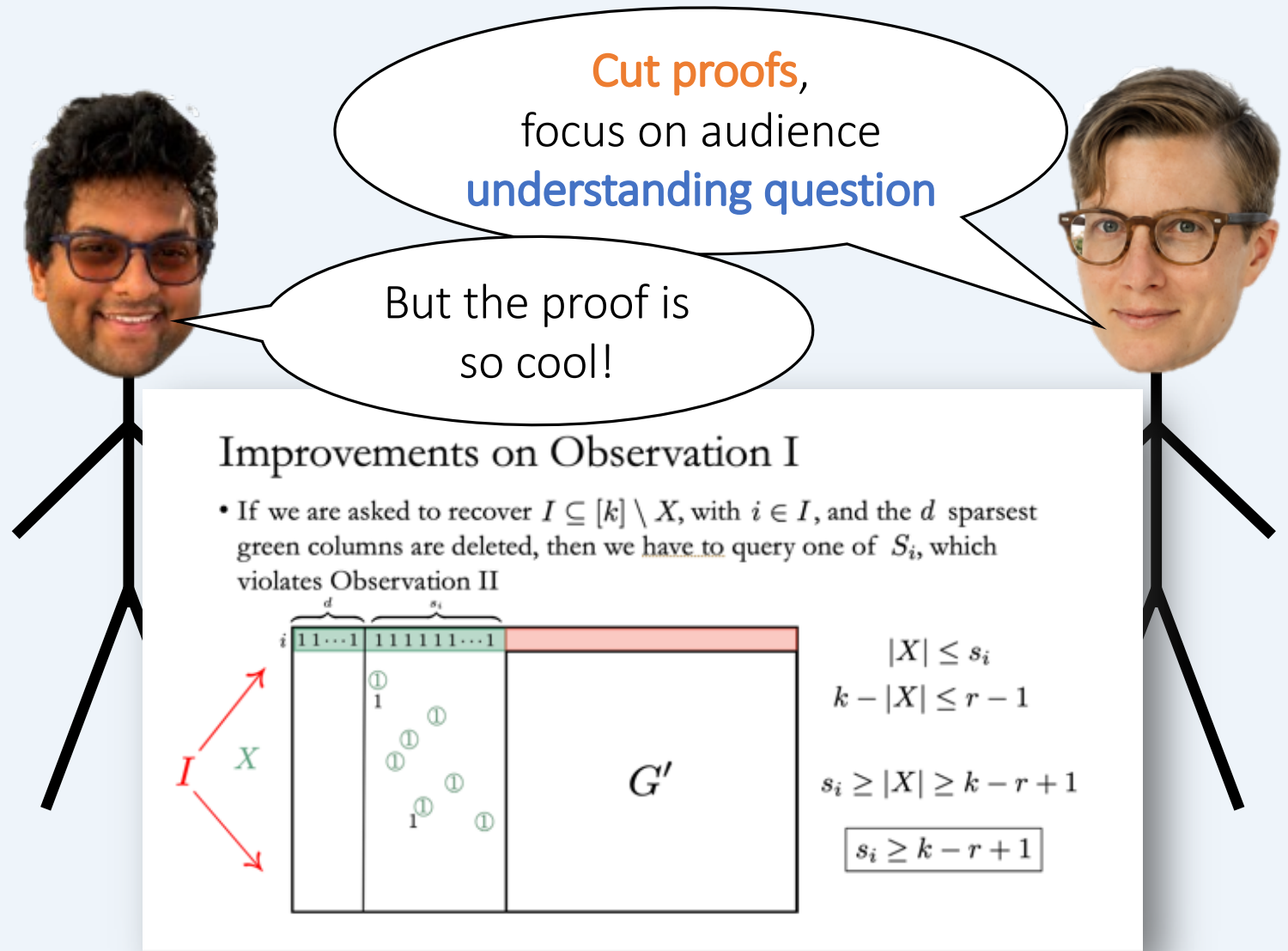
(credit to them blame to me)

I. Your Goal



My first talk

- My guess of **the goal**:
 - Set up **question**
 - State **theorem**
 - Explain **proof**

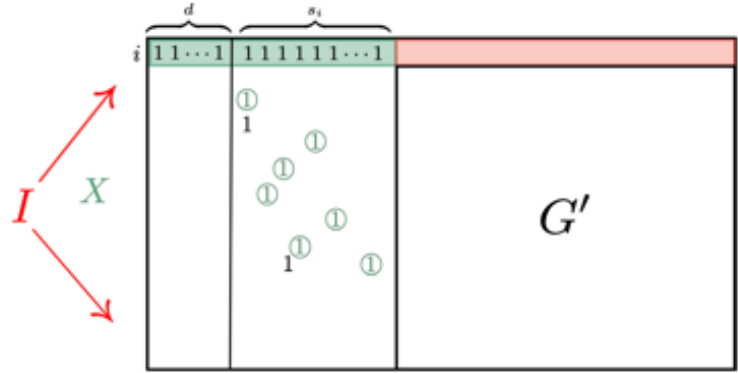


But the proof is so cool!

Cut proofs,
focus on audience
understanding question

Improvements on Observation I

- If we are asked to recover $I \subseteq [k] \setminus X$, with $i \in I$, and the d sparsest green columns are deleted, then we have to query one of S_i , which violates Observation II



$|X| \leq s_i$
 $k - |X| \leq r - 1$
 $s_i \geq |X| \geq k - r + 1$
 $s_i \geq k - r + 1$

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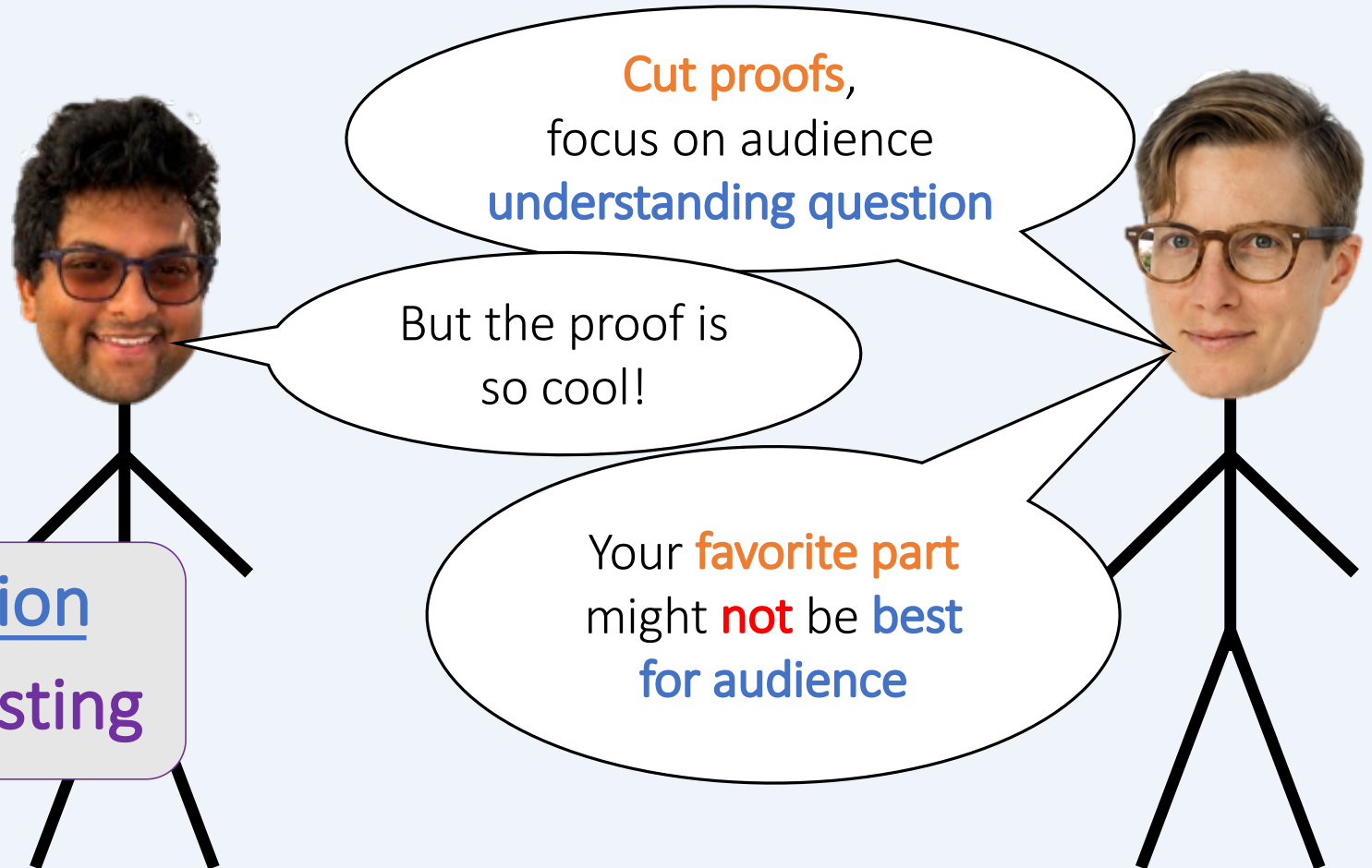
The Truth

Only goal: they understand question

Above and beyond: think it's **interesting**

Most talks do too much, become unintelligible

If everyone understands the question that's a great talk!



II. The Audience



Two Audience Archetypes

Curious Undergrad



Renowned Professor



Two Audience Archetypes

Curious Undergrad



Renowned Professor



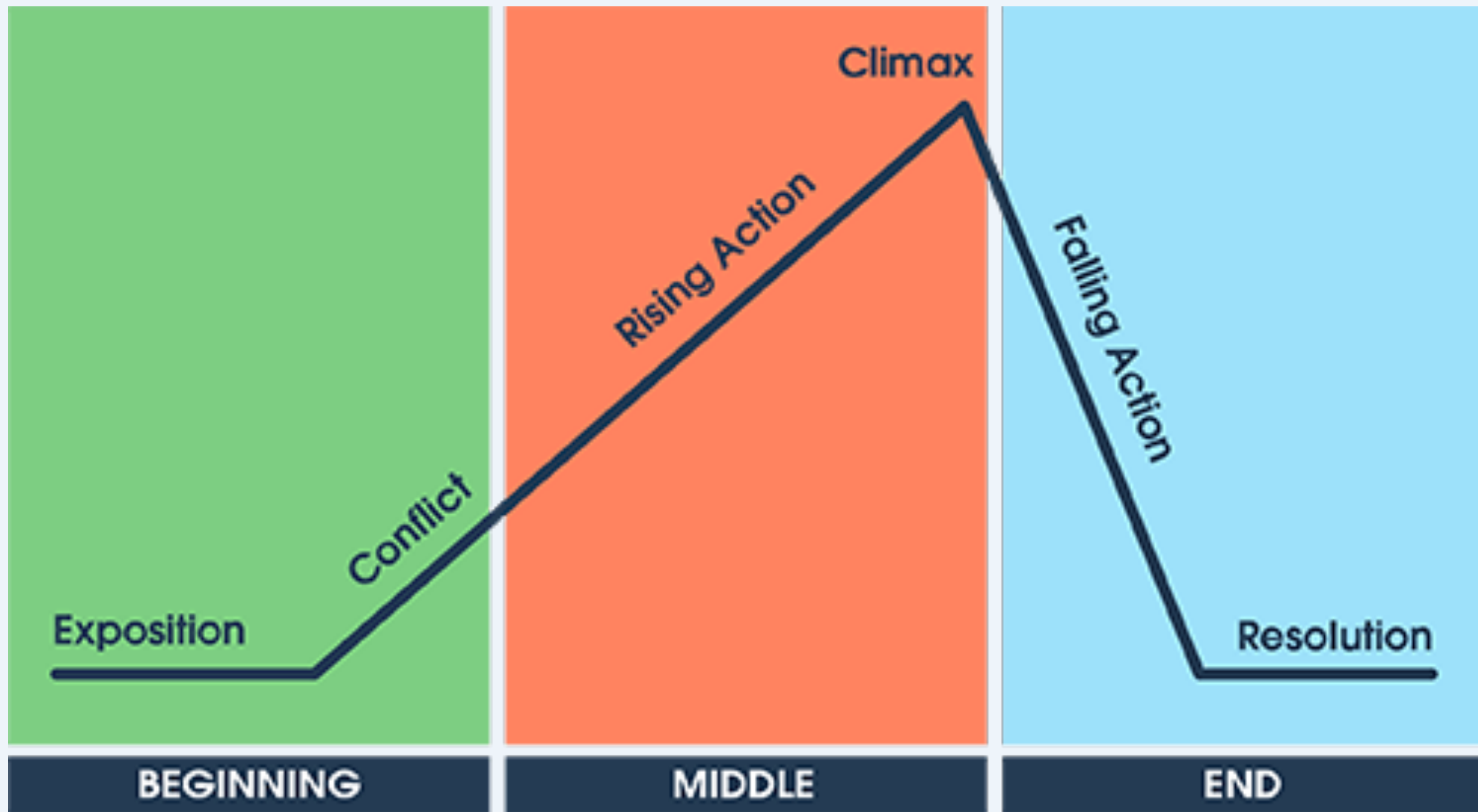
(answering emails)

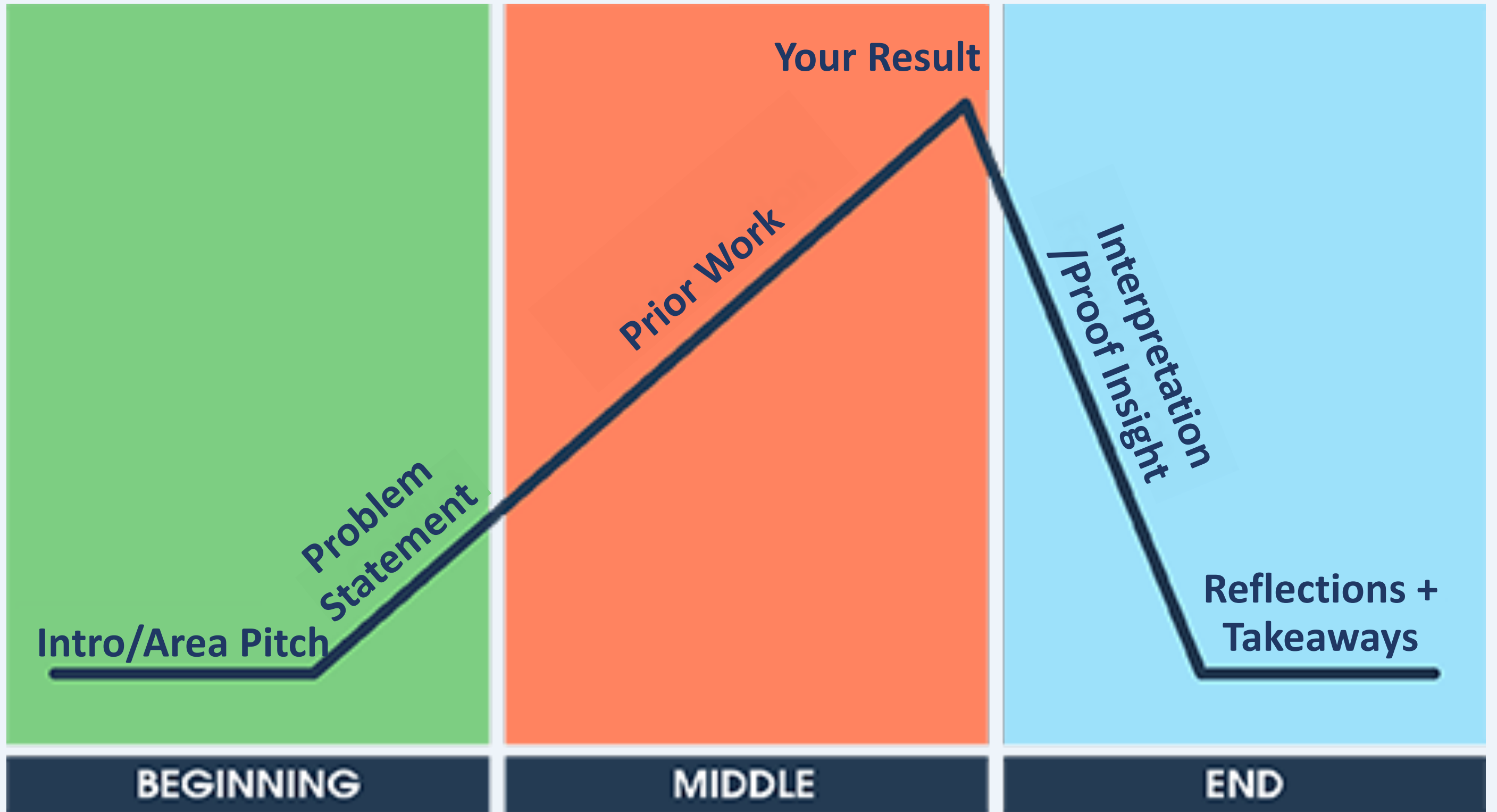
Hope: both understand everything

Tricks to Keep the Audience

- Keep it **simple** (avoid math), emphasize **big picture**
- **Repeat** everything important
 - Statement, contrapositive, concrete example
 - Formal + informal
- **Signpost aggressively**
 - “Here is the problem/result/takeaway”
- Make the talk **modular**
 - Map out dense sections, return to map
 - Give opportunities back in
- Create **dramatic tension, intrigue, anticipation**







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- Create **dramatic tension, intrigue, anticipation**
- Be **excited**, encourage **interaction!**



III. Interaction



Tricks for Interaction

- **Prime the audience**
 - Ask a question: “Can everyone hear me?”
 - **Invite interruptions**, enjoy them
- **Poll the audience**
 - “How many of you have seen...”
 - Invite guesses/ideas/discussion
- **Anticipate questions, lay traps**
 - Can bait questions by omitting details



IV. Slides

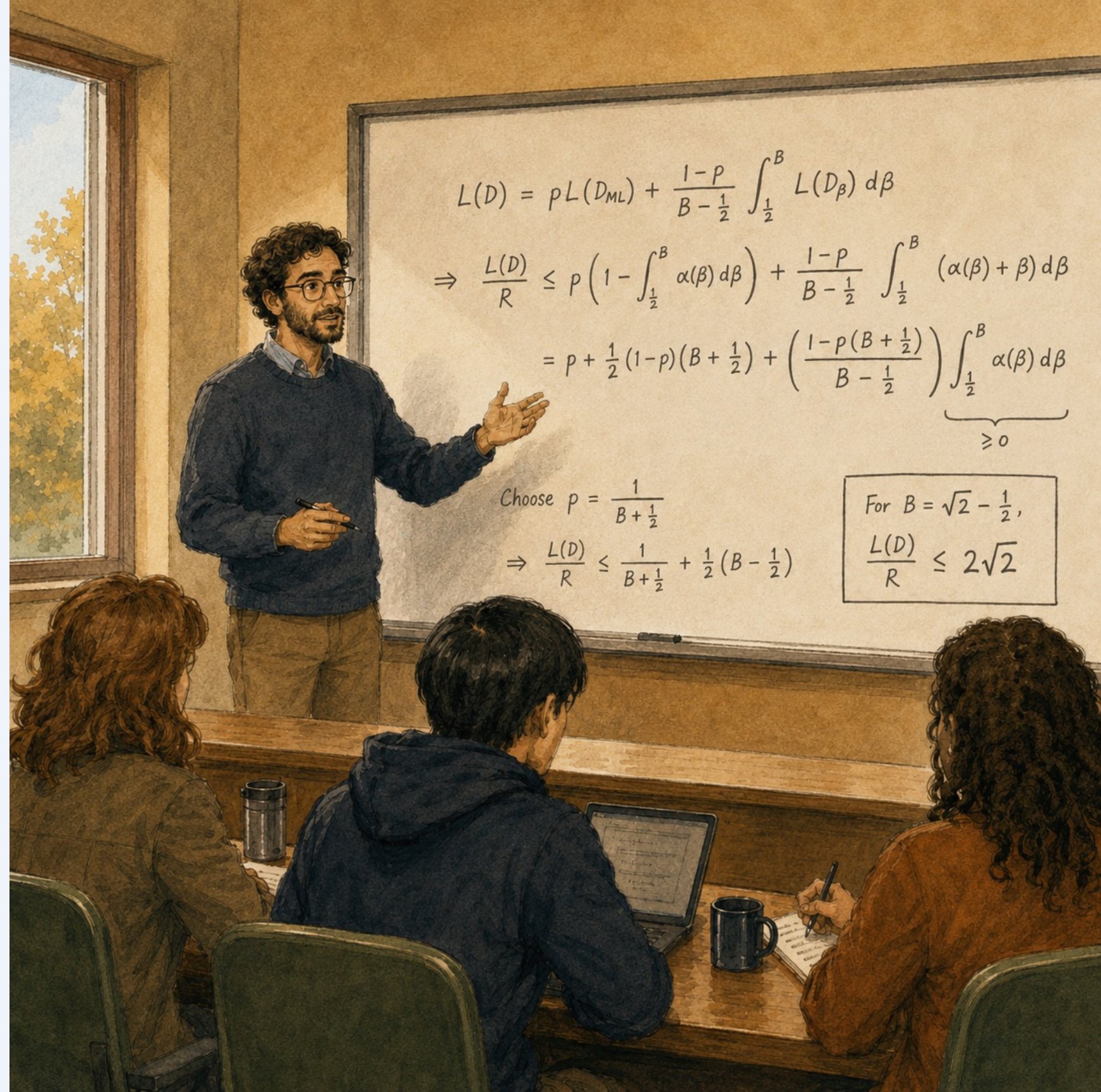


Tips for Slide Design

- Key principle: where do the **eyes** look? 🙄
 - Pictures, **colors**, boxes (anywhere but black text)
- Pictures + boxes – **trick audience** into **browsing slide**
- My approach: **colors** = **good summary** of slide
- **Minimize words** (brevity over grammar)
 - Often can remove articles
- **Avoid math**
 - (my) hot take: beamer talks *usually* worse

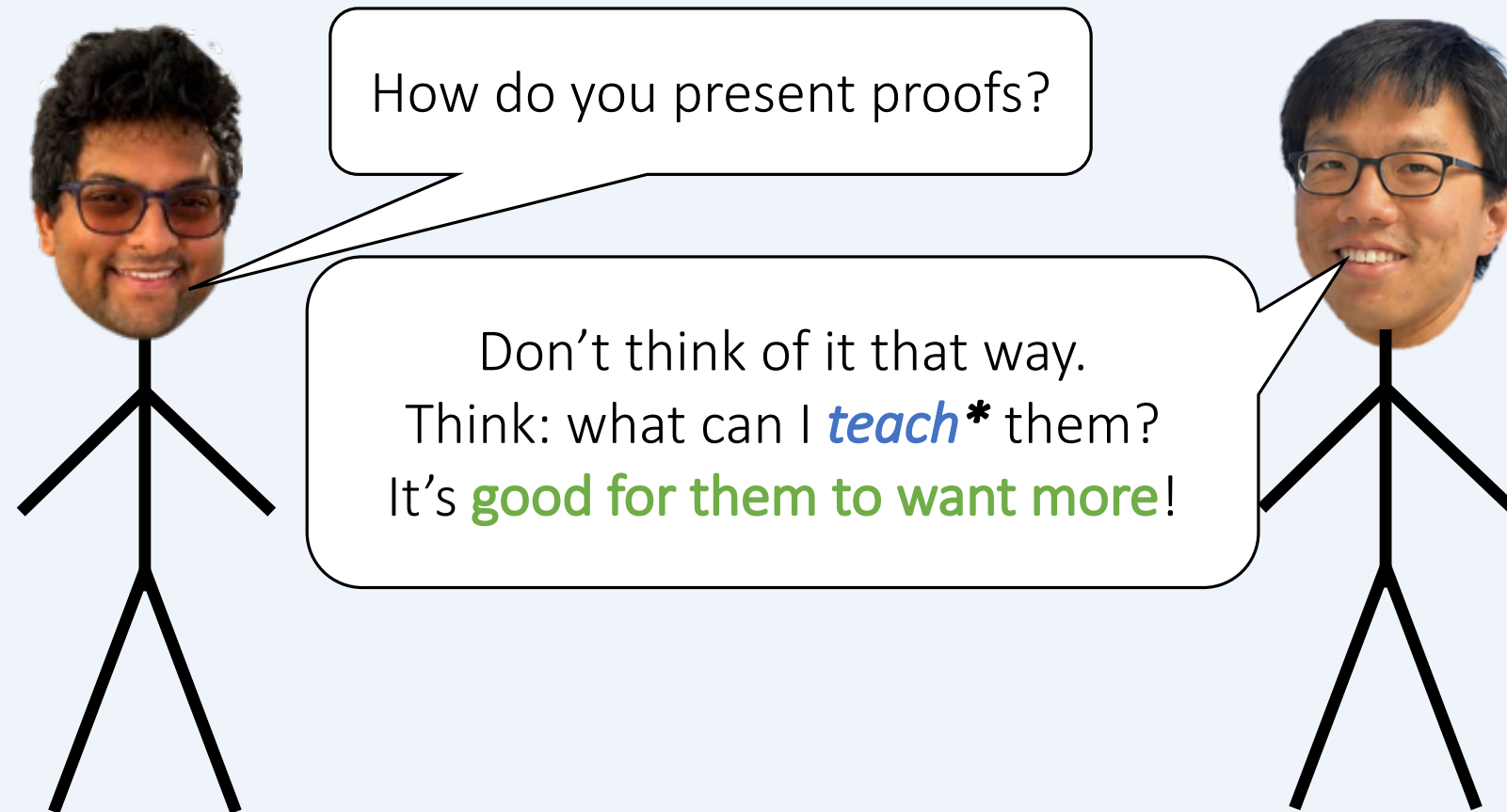


V. Proofs



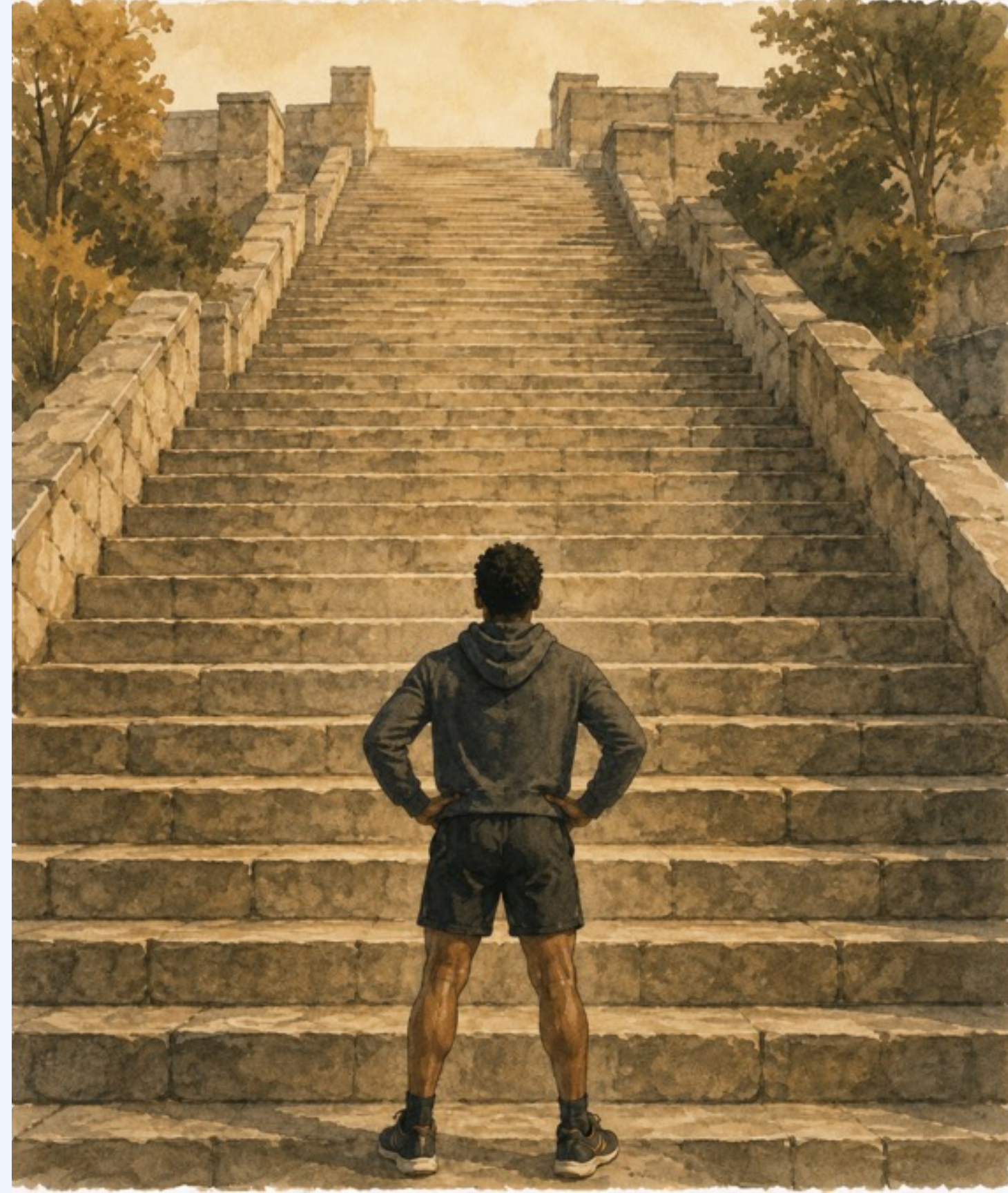
Proofs, if you must

- Long talks (1hr) may need proofs
- How to present proofs **without losing the audience?**



*e.g., **failed proof** + hint for **fix**, **useful technique**

VI. How to get better



Practice!

- Ideal: with a **mentor** or **friends**
- At home:
 - Keep track of **timing**, adjust accordingly
 - Helpful hack: if struggling to find words, **write a rough script** (not to use)
- Give a **talk many times**, iterate
 - **Seminars** locally, visiting friends...
- **Teach!** (HS students too)



Learn from Great Talks

- How'd they **keep your attention**?
- **Pace technical material**?
- What was **different from you**?
- What could you **borrow**?

Apr. 29, 2015



From Math Magic to Secure Communication

Dan Boneh
Professor
Stanford University



Intro to quantum codes

Disclaimer:

This scheme, like all other schemes for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work.

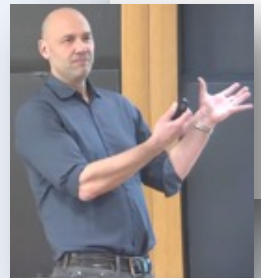
John Wright
UC Berkeley



Computer Science and Game Theory: A Conversation

Tim Roughgarden (Columbia University)

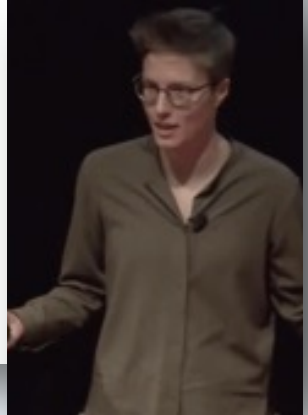
School of Mathematics Public Lecture
Institute for Advanced Study
November 19, 2025



Error Error Error
Correcting Correcting Correcting
Codes Codes Codes

Mary Wootters
Stanford

October 19, 2016



Optimal mean-based algorithms for trace reconstruction

Ryan O'Donnell
Carnegie Mellon

joint work with: Anindya De (Northwestern)
& Rocco Servedio (Columbia)



... and other talks

- Where did they lose you? Why?
- How could they have **brought you back in?**



Final Reflections

- Two parts of theory research:
 - Asking questions
 - Solving problems
- What if solving becomes cheap?
- Prioritize asking?
- Point of research: building understanding
- And communicating it!
- Prioritize communication too!



One more thing...

VII. Get a Clicker

Presentation Clicker with USB-A & USB-C Receiver, 2.4GHz Wireless Presenter Remote with Volume Control & Red Light Pointer, Clicker for PowerPoint Slides Slide Advancer for Mac Computer, Plug & Play

Brand: ELISWEEN

4.5 ★★★★★ (3,377) | [Search this page](#)

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2K+ bought in past month

-5% \$13⁶⁷ [Price history](#)

Typical price: \$14.39 ⓘ

[i](#) Last purchased Dec 28, 2023
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Lesson Cheat Sheet

- Main **goal**: they **understand the problem**
- The audience is **barely listening**
 - **Repeat, signpost**, give **ways back in**
 - Build **tension, intrigue, anticipation**
- **Invite interaction** (priming, polling), **anticipate questions, lay traps**
- Follow the 🙄 in slide design (**colors!**)
- In proofs (if you must), **aim to teach**
- **Practice, learn from other talks**

Thank you!

